NOT MEASUREMENT SENSITIVE

MIL-STD-361-9(TM)
15 JULY 1992

SUPERSEDING
(See 6.5.)

MILITARY STANDARD

MANUALS, TECHNICAL:

TROUBLESHOOTING PROCEDURES

AMSC A6758

AREA TMSS

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FOREWORD

1. This military standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DOD).

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander USAMC Materiel Readiness Support Activity, ATTN: AMXMD-EP, Lexington, Kentucky 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

3. This document is one part of an 11-part bookform standard to be used in the preparation and acquisition of technical manuals that are essential to support (operate and maintain) the various types of equipment and weapon systems within the Department of the Army.

4. This document covers the troubleshooting procedures requirements for the preparation of technical manuals.

5. Other parts of this multi-part standard are:

MIL-STD-361-1	-	Writing Style and Format Requirements.
MIL-STD-361-2	-	Comprehensibility Requirements.
MIL-STD-361-3	-	Requirements for Graphics.
MIL-STD-361-4	-	Quality Assurance Requirements.
MIL-STD-361-5	-	Requirements for Packaging.
MIL-STD-361-6	-	Introductory Information with Theory of Operations Requirements.
MIL-STD-361-7	-	Operator and Preventive Maintenance Instructions.
MIL-STD-361-8	-	Maintenance Instructions.
MIL-STD-361-10	-	Lubrication Instructions.
MIL-STD-361-11	-	Repair Parts and Special Tools List (RPSTL).

6. By incorporating standardized and service-unique technical manual requirements, this document and its dash-numbered companion parts, along with the general acquisition specification MIL-M-63008, Acquisition/Preparation Requirements, support the acquisition of technical manuals.

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1. SCOPE

1.1 Scope. This document establishes the content and procedural requirements for preparing troubleshooting instructions for technical manuals (TMs), revisions, supplements, and changes.

1.2 Purpose. This document supplements the various content requirements in the other parts to this 11-part standard for specific types of technical manuals and related publications and does not alone deliver any technical data. (NOTE: The figures used in this document are examples only. The text of this document takes precedence over the figures.)

1.3 Applicability. The applicability of the requirements contained in this document is governed by the equipment, systems, or weapon systems for which the technical manuals are being developed. This document is applicable for use by the Department of the Army.

1.4 Selective application and tailoring. This document contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in this document are the responsibility of the contracting activity and shall be accomplished through the use of MIL-M-63008. The applicability of some requirements is also designated by one of the following statements:

- unless requirement is specifically excluded by the contracting activity; or
- as/when specified by the contracting activity.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2.)

SPECIFICATIONS

MILITARY

MIL-M-63008	-	Manuals, Technical: Acquisition/ Preparation Requirements.

STANDARDS

MILITARY

MIL-STD-12	-	Abbreviations for Use on Drawings, and in Specifications, Standards, and Technical Documents.
DOD-STD-100	-	Engineering Drawing Practices.
MIL-STD-361-1	-	Manuals, Technical: Writing Style and Format Requirements.
MIL-STD-361-2	-	Manuals, Technical: Comprehensibility Requirements.
MIL-STD-361-3	-	Manuals, Technical: Requirements for Graphics.
MIL-STD-361-4	-	Manuals, Technical: Quality Assurance Requirements.
MIL-STD-361-5	-	Manuals, Technical: Requirements for Packaging.
MIL-STD-361-6	-	Manuals, Technical: Introductory Information with Theory of Operations Requirements.
MIL-STD-361-8	-	Manuals, Technical: Maintenance Instructions.
MIL-STD-1388-2	-	DOD Requirements for a Logistic Support Analysis Record.

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

2.1.2 Other Government document(s) and publication(s). The following other Government document/publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

TB 750-93-1 - Functional Grouping Codes: Combat, Tactical, and Support Vehicles and Special Purpose Equipment.

(Copies of other Government documents and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document ("This document" includes all volumes of MIL-STD-361 series.) and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

a. BIT - Built-In Test.

b. BITE - Built-In Test Equipment.

c. DOD - Department of Defense.

d. DODISS - Department of Defense Index of Specifications and Standards.

e. FGC - Functional Group Code.

f. FMC - Fully Mission Capable.

g. LSA - Logistic Support Analysis.

h. LSAR - Logistic Support Analysis Record.

i. MAC - Maintenance Allocation Chart.

j. MAPS - Maintenance Action Precise Symptom.

k. RAM - Reliability, Availability, Maintainability.

l. SI - System Integration.

m. SMR - Source, Maintenance, Recoverability.

n. TM - Technical Manual.

3.2 Assembly. Two or more parts or subassemblies joined together to perform a specific function and capable of disassembly (e.g., power shovel-front, fan assembly, audio frequency amplifier).

NOTE

The distinction between an assembly and subassembly is determined by the individual application. An assembly in one instance may be a subassembly in another where it forms a portion of an assembly.

3.3 Component. A constituent part not normally considered to be capable of independent operation; a piece part.

3.4 Contracting activity. The Department of Defense (DOD) component, activity, or organization of a using military service, or that organization delegated by a using service, that is responsible for the selection and determination of requirements for TMs.

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3.5 Equipment. One or more units capable of performing specified functions.

3.6 Equipment, noncomplex. Equipment with relatively simple systems and on-board equipment installed; especially with a minimum of sophisticated electronic systems, integrated systems, and computer controlled built-in test (BIT) capability. Equipment in this category requires a low volume of technical information to adequately support operation and maintenance. Installed systems, subsystems, and on-board equipment, support equipment, and spare parts can be effectively consolidated into a small number of highly usable documents.

3.7 Fully mission capable (FMC). Equipment that can perform all of its combat missions without further endangering the lives of crew or operators in a combat environment.

3.8 Maintenance action precise symptom (MAPS). Troubleshooting symptom precisely described which has troubleshooting procedure(s) found on a MAPS list.

3.9 Repair part. Any nonreparable component (Supply Class IX item) required for maintenance/repair of an end item/ equipment (will be coded Z or B in fourth position of the source, maintenance, recoverability (SMR) code).

3.10 Replace. To remove an unserviceable spare or repair part and install a serviceable counterpart in its place. Replace is authorized by the LSA/MAC and the assigned maintenance level is shown as the third position code of the SMR code.

3.11 Schematic diagram. A graphic representation showing the interrelationship of each component or group of components in the equipment. The essential characteristic of these diagrams is that every maintenance-significant functional component is separately represented.

3.12 Spare part. Any reparable and recoverable component required for the maintenance or repair of an end item or equipment (will have a recoverability code other than "Z" and will be assigned an FGC or subfunctional group code in the MAC).

3.13 Subsystem. A subordinate combination of complete operating equipment that performs a specific function within a system.

3.14 Support equipment. Equipment necessary to maintain, operate, and/or support the system or subsystem covered.

3.15 System. A regularly interacting or interdependent group of items forming a unified whole to perform a specific operational function.

3.16 Technical manuals (TMs). Manuals that contain instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement

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between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual displays, electronic imbedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures.

3.17 Unit. An assembly or any combination of spare and repair parts mounted together, normally capable of independent operation in a variety of situations.

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4. GENERAL REQUIREMENTS

4.1 General style requirements.

4.1.1 Conformance to standard. The general manner for the preparation of the cover page, front matter, text, security classification, classification markings, numbering, changes, warnings, cautions, health hazards, notes, final reproducible copy of final TM, and other document elements pertaining to troubleshooting procedures shall be in accordance with MIL-STD-361-1.

4.1.2 Technical content preparation. The following baseline requirements shall be used as specified by the contracting activity:

a. Logistic support analysis record (LSAR). The technical data and instructions developed by the requirements of MIL-STD-1388-2, DOD Requirements for a Logistic Support Analysis Record (LSAR), shall be used as the baseline to prepare TMs.

b. Maintenance allocation chart (MAC). The MAC shall be used as the baseline to prepare TMs. The MAC shall be prepared from the equipment top-down breakdown to consolidate and identify those groups on the list which involve identified maintenance functions. LSA Output Report 04, Maintenance Allocation Summary will be used when available as source data for the final approved MAC.

When specified by the contracting activity and if a waiver of standard requirements is granted, TB 750-93-1 may be used as the baseline for preparing TMs.

4.1.3 Security classification markings. All classified information and material shall be safeguarded and identified in accordance with MIL-STD-361-1.

4.2 Contents. Troubleshooting procedures shall begin with a failure fault symptom and diagnose to a single fault/failure. Troubleshooting shall refer to specific maintenance or repair tasks to correct the fault. The troubleshooting instructions shall provide all essential information required by user personnel for performing all required troubleshooting on equipment, systems, and weapon systems. Troubleshooting instructions shall include detailed inspection and troubleshooting information (including functional descriptions of subsystems being diagnosed) to enable the user of the TM to troubleshoot and return to an acceptable performance standard all components of the equipment in a minimum of time with the skills, tools, test equipment, and spare parts authorized in the MAC and RPSTL or as defined by the LSA. Instructions shall cover all items comprising the equipment, such as accessories, components, wiring, and junction boxes. Information and the necessary remedy shall be included. The method used for identifying system

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equipment test points to include the requirements and methods of determining defects through visual inspection shall be explained. References to information contained elsewhere shall be in accordance with MIL-STD-361-1. Instructions, where applicable, will flow from operator level through unit, direct support, and general support until fault is isolated. Procedures will include schematics and drawings as needed. Fault symptoms must be the same from lowest level through highest level, until the diagnosis is complete.

4.2.1 Development of text and graphics. Applicable requirements of MIL-STD-361-1 and MIL-STD-361-3 shall apply when developing text and graphics. References to information contained elsewhere shall be in accordance with MIL-STD-361-1. Emphasis shall be placed on the specific steps of the troubleshooting procedures to be followed, the results that may be expected or desired, and the corrective measures required when such results are not obtained. Graphics shall include locator art, test connectors, test probes, etc., as necessary for clarification.

4.2.2 Environmental control data. Environmental control data provided by the contracting activity shall be incorporated. The instructions shall include information on any special troubleshooting procedures required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment.

4.2.3 Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-STD-361-1.

4.2.4 Abbreviations. Abbreviations shall be in accordance with MIL-STD-361-1 and MIL-STD-12.

4.2.5 Reference designations. Electrical and electronic reference designations shall be in accordance with DOD-STD-100.

4.2.6 Cable fabrication. Refer to MIL-STD-361-8.

4.2.7 Measurements and tolerances. All numerical data shall be presented in accordance with MIL-STD-361-1.

4.2.8 Safety devices and interlocks. The purpose and location of all safety devices and interlocks shall be given in conjunction with the pertinent procedures.

4.2.9 Use of warnings, cautions, and notes. Refer to MIL-STD-361-1.

4.2.10 Use of color. When approved by the contracting activity, use of color shall be in accordance with MIL-STD-361-3.

4.2.11 Glossary. If the TM contains a number of terms requiring definition, a glossary shall be included in accordance with MIL-STD-361-1.

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4.2.12 Advertising. Except for the identity of the equipment manufacturer or contractor, publication material prepared in accordance with this standard shall contain no advertising matter.

4.2.13 Final reproducible copy. Preparation of final reproducible copy shall adhere to the requirements specified in MIL-STD-361-1.

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5. DETAILED REQUIREMENTS

5.1 General. TM troubleshooting procedures prepared in accordance with this document shall be as specified by the detailed requirements herein. The types of troubleshooting procedures and the types of TMs or publications wherein the procedures are to appear shall be as specified by the contracting activity. For systems with system integration (SI) manuals, troubleshooting for any fault shall begin in the SI manual. Troubleshooting shall continue in the SI manual until the fault or symptom can be isolated to a specific component. At this point, further troubleshooting shall be referenced to the system component TM in such a manner that the correct troubleshooting path in the system component TM is easily recognized. The information shall be presented by simplified illustrations, charts, tables, or diagrams in the format most suitable for the equipment. Writing style and format shall be in accordance with MIL-STD-361-1 and MIL-STD-361-2.

5.1.1 Coverage in a single TM. When operations, maintenance, and troubleshooting procedures are covered in a combined single TM, the technical content shall be at the same detail level as if covered in several specific TMs. Troubleshooting shall be in the applicable maintenance chapter as determined by level of maintenance. For arrangement of such a combined single TM, refer to MIL-STD-361-8.

5.1.2 Applicability of troubleshooting to the various levels of maintenance. TM troubleshooting data shall promote fault isolation in equipment that will allow the technician to quickly remedy the problem and return the equipment to operational status. The data shall address the applicable maintenance level. Procedures must flow from operator's TM to unit, direct support, and general support maintenance TMs, as applicable, to isolate fault. Troubleshooting procedures shall refer to the applicable maintenance procedure after fault isolation. Instructions to follow all repair, replace, or adjust actions resulting from troubleshooting with a repair verification check to help detect multiple faults and to ensure that the repair, replace, or adjust action cleared the fault shall be included.

5.2 Preparation of TM troubleshooting data.

5.2.1 Text content. Collectively, the TMs shall contain all information required to troubleshoot and support the equipment for which these TMs are prepared. Sound engineering principles and techniques, LSA, engineering analysis, service experience, performance data on similar items, and all other reliability, availability, maintainability (RAM) data available shall be used in the preparation of specific TMs.

5.2.2 Use of MAC. As applicable, TM troubleshooting data prepared in accordance with this standard shall describe, in appropriate detail, the troubleshooting procedures authorized by the MAC. See MIL-STD-361-8.

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5.3 Troubleshooting. This section shall contain those procedures that will isolate defects to the level authorized by SMR coding remove/replace and shall be prepared in the following manner.

5.3.1 Troubleshooting data. Troubleshooting TMs, chapters, or sections shall contain procedures essential for troubleshooting equipment. Data shall be consistent with the capabilities of the intended user and use of common tools, test equipment, and repair parts as authorized by the LSA and/or MAC and SMR codes assigned to the item. Troubleshooting data should be test and fault-isolation oriented. Instructions to repair by replacing parts until the problem disappears is to be avoided if at all possible.

5.3.2 Troubleshooting procedures. Troubleshooting procedures shall provide instructions for quick fault-isolation of defects in equipment. The procedures shall be presented so the user moves directly from the symptoms of the problem to the proposed corrective actions with a minimum of information and movement and will allow the user to quickly identify, repair, and return the equipment to a fully mission capable status condition in minimum time. Factors that shall be considered in the development of troubleshooting procedures are as follows:

a. Maintenance concept.

b. Ease of testing.

c. Test access time.

d. Component reliability.

e. Test time.

f. Automated versus manual testing.

g. Test equipment requirements and availability.

h. User environment.

i. Target audience.

j. System quick-turnaround requirements.

5.3.3 Major troubleshooting divisions. Troubleshooting shall be divided into three major steps: fault discovery, troubleshooting data entry, and troubleshooting accomplishment. Each of these steps has alternatives that are discussed in detail in 5.3.5 through 5.3.8. They include:

a. Fault discovery:

(1) By operator.

(2) During scheduled or on-condition maintenance by maintenance personnel.

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b. Troubleshooting data entry:

(1) During operation - go to maintenance action precise symptom (MAPS) list.

(2) During maintenance -

(a) Include in maintenance procedures.

(b) Refer to troubleshooting procedures.

(c) Go to MAPS list.

c. Troubleshooting methods:

(1) Logic trees.

(2) Troubleshooting diagrams.

5.3.4 Troubleshooting and fault-isolation requirements. Step-by-step, proceduralized troubleshooting shall be included.

5.3.4.1 Malfunctions covered by logic trees. Logic trees, when used in conjunction with any "built-in" test or troubleshooting aids, shall with a high degree of accuracy lead the troubleshooter to faulty replaceable parts. Logic trees shall include as many malfunctions as possible that actually occur during equipment operation or during normal maintenance.

5.3.4.2 Malfunctions covered by troubleshooting diagrams. For those malfunctions that are not covered by logic trees, the TM user shall be directed to a troubleshooting aid (i.e., troubleshooting diagrams) with which further fault analysis of available information is possible.

5.3.5 Fault discovery. Malfunctions in a system are discovered in one of two ways:

a. The symptoms of a fault occur during normal operation.

b. The fault shows up during normal maintenance such as inspection, alignment, check procedures, or scheduled testing/checkout.

5.3.6 Troubleshooting data entry - trouble discovered during operation. When a malfunction is discovered during equipment operation, the entry into the troubleshooting procedure is based on the actual symptom. The maintenance technician should be directed to debrief the operator for a general description of the problem and should then be instructed to attempt to verify the fault. In so doing, the technician may or may not observe the same fault symptom. If the same fault symptom is observed, the technician should be directed to attempt to define the symptom as accurately as possible. The malfunction description is then used to refer to the MAPS list - a list that correlates the symptoms to precise troubleshooting procedures.

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5.3.6.1 MAPS list requirements. The MAPS list (figures 1 and 2) is required for all operation-discovered-fault troubleshooting data entries. It must meet the following requirements:

a. Contain fault symptoms for all known malfunctions.

b. For complex systems, list symptoms by subsystem categories, if necessary, and may use codes that help identify specific items.

c. Catalog symptoms by method of detection if this aids usability.

5.3.6.2 MAPS list preparation. The MAPS list can be prepared using either the tabular or branching method. The symptom descriptions that make up the list are derived from those formulated from the logic trees and are identical. For medium-complex systems, either method is acceptable provided the list contains the minimum data of precise symptom description and troubleshooting procedures reference. For simple systems, figure 1 gives an example of a tabular method that could be used. Figure 2 gives an example of a method which may be used for medium- to heavy-complex systems.

5.3.7 Troubleshooting data entry - trouble discovered during equipment maintenance. When the fault is discovered during scheduled/on-condition maintenance, three choices exist for entry into the troubleshooting data. These alternatives are listed in order of preference. Specific requirements, advantages, and limitations for each of these follow.

5.3.7.1 Troubleshooting integrated into scheduled or on-condition maintenance procedures. See figure 3.

a. Designed so technician does not have to shift to other part(s) of TM.

b. Suitable for simple- to medium-complex systems.

c. Should be used whenever volume of data permits.

5.3.7.2 User referred to separate troubleshooting data. See figures 4 and 5.

a. Refers technician directly to proper troubleshooting data at each step of procedures where a fault could be encountered.

b. Refers to troubleshooting aids for complex/unusual faults.

5.3.7.3 User referred to MAPS list by symptom.

a. Refers to MAPS list either at beginning of procedure or at point where malfunction could be encountered.

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b. May use MAPS entry code in lieu of troubleshooting reference.

c. Requires additional searching/decision making by technician.

5.3.8 Troubleshooting methods. The following methods shall be used, as applicable, in the preparation of troubleshooting data:

a. Logic trees.

(1) Tabular.

(2) Branching.

(3) Narrative.

b. Troubleshooting diagrams.

Whichever method is used shall require performance of any applicable self-tests, alignments, and inspections before accomplishing other troubleshooting.

5.3.8.1 Logic trees. Logic trees, proceduralized troubleshooting format is a programmed method that involves user interaction. The user shall be given a particular instruction, then asked a yes-no question about the result of the action. Based on the answer, the user shall be referenced to another instruction. All troubleshooting employs some variation of this method. The logic trees shall be based on failure symptoms and include the most probable faults with the most probable causes. Logic trees shall contain all essential and pertinent information that would be included in any other form of procedure. This includes warnings, cautions, notes, power turn-on procedures, precheckout procedures, reference diagrams, initial switch settings, and task setup frame information. Logic trees shall be prepared assuming one malfunction at a time is being corrected. The first steps in the logic trees shall verify the reported failure actually exists and the proper logic tree is being used. An operational check shall be specified to be performed after the fault is corrected to ensure correct operation of the system. The functional description at the beginning of the logic tree procedure shall describe functioning of the system/subsystem being diagnosed.

5.3.8.1.1 Logic tree preparation. The logic tree shall provide the maintenance technician with a means of isolating system failures to the faulty line replaceable item/shop replaceable unit. The depth of coverage in the logic tree may range from a unit to a subassembly. To prepare the troubleshooting logic tree, exercise the following general rules:

a. Study the system in detail using all available technical documents, such as schematic diagrams, test specifications, design descriptions, and trouble/failure reports. Understand the system operation and note what

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functions are performed by the units associated with the system.

b. Perform a fault analysis on paper, listing all problem areas and faults that could possibly occur. Because fault isolation must be accomplished in the least amount of time, the logic tree shall begin with tests of components with the least reliability or tests which require the least time to be performed.

c. After determining the correct output, check toward the source until fault is located.

d. Develop the logic tree based on tests, measurements, and decisions that must be made in order to reach the final outcome of isolating the fault to a replaceable component related to the symptom. All of the most probable faults and causes shall be considered.

e. Include any information that will aid the technician, such as waveforms, voltage levels, references to test diagrams, functional diagrams, test, etc., and alignment procedures, checkout procedures, or other scheduled maintenance procedures. Include realistic values and tolerances, locator art, test points, connector ends, etc. The end result must be the repair of the system.

f. List any self-tests that are associated with the system. Self-test schemes shall be the prime troubleshooting tool, with manual troubleshooting by logic tree taking over where self-test leaves off or fails to locate malfunction. Build the procedure using system self-tests before using external test equipment. Avoid procedures that use excessive test equipment or require measurements to be made in locations difficult to access.

g. Ensure that actions required are reasonable to perform. For example, do not ask the technician to connect a meter at one point and set a switch at another point where the meter reading cannot be observed. Be practical.

h. Validate proper operation of a rough draft procedure by the most efficient and practical method - preferably on the actual hardware.

i. Whenever possible, create the fault conditions on the system by inserting faults and verifying results and corrective actions.

j. When all possible faults have been determined and described, prepare the MAPS list using the exact description of the fault as the symptom description. Group symptoms to common system areas. For example, if a system has a data link, communications, radar, display, and tracking systems, the symptoms would be grouped into each related area. All fault symptoms of a communications

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nature would fall into the communications group. The symptoms could be further divided into functions within the communications group that would be common. The same would be done for radar, data link, display, and tracking systems.

k. Ensure that the logic tree contains all the required references to other troubleshooting aids and that all references are correct.

5.3.8.1.2 Use of logic trees. Logic trees shall be used, as applicable, to allow greater flexibility for the isolation of faults and improved conceptualization of the system by the technician.

5.3.8.1.3 Logic tree format. Three formats for logic tree troubleshooting procedures are shown in figures 6 through 12 and their advantages and limitations are listed in Table I. As applicable, the logic tree format shall consist of the following:

a. Location.

b. Item or indicator.

c. Action.

d. Indication.

e. Corrective action.

(1) If indication is obtained.

(2) If indication is not obtained.

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f. Remarks.

TABLE I. Logic tree formats.

Logic Tree Format	Advantages	Disadvantages
Tabular (fig. 6 and 7)	Compact; on one page Can be produced on preprinted forms Relatively easy to use	Procedure flow may not be easily seen
Branching (fig. 8, 9, and 10)	Procedure flow readily apparent Suitable for computer generation	Requires more space than tabular
Narrative (fig. 11 and 12)	Easy to write and produce	Requires more time and is tedious to use Procedure flow not evident

5.3.8.2 Troubleshooting diagrams. Troubleshooting diagrams can be used to supplement logic tree troubleshooting data or as stand-alone troubleshooting aids. The three types of diagrams commonly used for troubleshooting purposes are system functional diagrams, test-loop functional diagrams, and fault isolation diagrams. For troubleshooting diagrams to be used as stand-alone aids requires a highly experienced technician with considerable system experience.

5.3.8.2.1 Troubleshooting diagram general requirements. Text shall explain diagrams to provide maintenance personnel with a description of the diagram in an overall terse manner. Enough material shall be presented to allow an understanding of what the troubleshooting is trying to accomplish and what the expected result(s) will be. The diagram shall show the operation of the equipment, systems, and subsystems as an operational unit. Relationships from one system to another and wiring from one component to another at the system and subsystem levels shall be shown. When required for clarity, wire numbers shall be used on the diagram. Input and output signals between subsystems, controls, and indicators shall be included. Test points and instrument hook-up shall be shown, noting the tolerance tests of all signals. Waveforms (including scope if applicable) shall be shown. Control indicators referred to in troubleshooting procedures and mechanical linkage shall be shown as required.

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a. System function diagrams:

(1) Depict functional operation of system.

(2) Show system broken down by functions.

b. Test-loop functional diagrams (figure 13):

(1) Show all circuits involved in test loop.

(2) Are stimuli and response oriented.

(3) Show built-in or automatic test provisions.

(4) Include all test conditions: test stimuli, component under test, and test measurement equipment.

c. Fault-isolation diagrams:

(1) Show all circuits involved in troubleshooting loop.

(2) Are symptom- or indication-oriented.

(3) Show same information as test-loop functional diagrams.

5.3.8.2.2 Troubleshooting diagram preparation instructions. Troubleshooting diagrams (schematic and functional) shall conform to the following principles:

a. Properly organized layout:

(1) Clearly traceable signal flow patterns.

(2) Uncluttered, uncrowded appearance.

(3) Minimum of crossed lines.

(4) Sufficient spacing for legibility.

b. Consistent:

(1) Standard referencing system for associated text, signal flow, and other diagrams.

(2) Standard names for hardware (name and reference designator), functions, signals, controls, indicators, and switches.

(3) Standard graphic symbols.

c. Complete detail:

(1) Hardware.

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(2) Function.

(3) Signal identification.

(4) Measurement data (voltage and waveforms).

(5) Explanatory text.

(6) Connectors.

(7) Terminal boards.

(8) Pin numbers.

(9) Signal names.

(10) Reference designators.

(11) Component values and tolerances.

d. Pertinent detail:

(1) Information not pertinent to purpose of diagrams omitted.

(2) Details of nonreparable and nonreplaceable components minimized.

5.3.9 Troubleshooting method selection criteria for TMs.

5.3.9.1 Testing and troubleshooting. Practical troubleshooting and operational checks shall guide the user to the cause and correction of equipment failure. All information required to perform these tests and evaluations shall be incorporated for all modes of operation. Troubleshooting diagrams shall be prepared as backup information in support of the troubleshooting procedures. The diagrams shall be referenced in the applicable troubleshooting text. Procedural information shall be based on the following assumptions:

a. Testing and troubleshooting procedures include the pretest setup (figures 14 and 15) and post-test teardown.

b. Procedures normally trace one malfunction at a time. However, possible multiple failures should be anticipated and considered in the testing procedures.

c. Successful completion of an operational checkout verifies system operation.

d. When troubleshooting procedures are not successful, references must be provided to troubleshooting diagrams (figure 16) in order that troubleshooting of unforeseen problems can be accomplished.

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e. If several items or components are suspected as the problem area, the one most likely to have failed shall be considered first; however, consideration shall be given to cost, access and ease of replacement when the choice is not clear. (See figures 17 and 18 for fault logic text, procedural text, examples, and tables for troubleshooting.)

5.3.9.2 Testing and troubleshooting for system integration (SI) manuals. Integrated complex system (e.g., aircraft) troubleshooting procedures shall be presented in two levels: first, the integrated system as a unit, and second, the systems that make up the integrated system. At the integrated system level, troubleshooting shall consist of a set of procedures leading to a visual or audible indication, readout, or display. The logic tree diagrams shall contain information on the exact branching of a given system channel that will permit troubleshooting a problem by its symptoms. For example, if two displays are abnormal, then the problem is in the portion of the channel common to both. When a test is called for and the result is obtained, the process shall continue through a new series of instructions leading to a new indication. If the sequence produces all desired results, it shall be assumed that the system or subsystem is operating satisfactorily. If any result called for cannot be obtained, the fault isolation shall consist of further instructions that will lead to a defective subsystem and, if possible, to an item, component, or interface wire. All known malfunctions that might prevent successful pre-operation, operation, or post-operation checks shall be considered, including failure of BIT/built-in test equipment (BITE). Following the integrated system, the associated systems shall be presented in system and subsystem order, as applicable.

5.3.9.3 Maintenance - testing and troubleshooting. The environment in which testing and troubleshooting procedures will be performed and the volume (bulk) of information required shall be considered. Testing and troubleshooting for a given system shall be presented in one chapter or in a series of chapters either within a system TM or in a separate TM, as specified by the contracting activity. Testing and troubleshooting procedures shall be prepared at the user level for the specific purpose of attaining rapid and effective fault isolation to a defective component or equipment. Following removal and replacement of a defective unit, the same procedures or a portion of these procedures shall be used to ensure system integrity and fully mission capable status.

5.3.9.4 Testing and troubleshooting procedures using nonautomated test equipment. Testing and troubleshooting procedures using nonautomated test equipment shall be established on the basis of a system test concept. To reduce maintenance downtime and decrease fault detection time, malfunction symptoms shall identify specific ports of entry into the testing and troubleshooting cycle. Repetition of time-consuming end-to-end tests shall be avoided.

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5.3.9.4.1 Procedural content using nonautomated test equipment. Procedures shall include but not be limited to the following:

a. An identification of the system under test.

b. A concise explanation of the testing and troubleshooting format.

c. A descriptive table identifying components to be installed and an identification of the access to each. The table shall be accompanied by an illustration that locates each component, equipment, accessory, connector, or junction box in the system under test. The table shall also identify every test connector or other test point to be used in the test.

d. A list of support equipment required for conducting the test.

e. Step-by-step pretest setup procedures that include the setup and use of support equipment, as applicable.

f. An illustration of all controls and displays, as required. (See figure 19.)

g. A complete table of test options shall be stipulated by fault logic text or logic tree format. In the case of simple noncomplex mechanical equipment, the use of fault and probable cause troubleshooting tables is acceptable. However, a clearly defined path must be evident from the fault description to its probable cause and correction. (See figures 14, 15, 16, and 18.)

h. Complete step-by-step test and troubleshooting procedures, including instructions required for use and application of installed on-line testing equipment. Procedures shall take into account controls, test point accessibility, indicators, and the feasibility of using BIT and automated on-board checkout equipment where available.

i. Identify connector numbers, pin designations, voltages, waveforms, resistance information, fluid pressures, and additional data required by the technician to perform the check and arrive at the correct diagnosis.

j. General safety precautions applicable to testing and troubleshooting.

5.3.9.4.2 Troubleshooting procedures using semi-automatic and automatic test equipment. Troubleshooting procedures using semi-automatic and automatic test equipment shall incorporate the same basic test information as described in 5.3.9.4.1, with the following exceptions:

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a. Testing setup procedures and troubleshooting criteria contained in separate TMs shall not be duplicated, but referenced.

b. Reference should be made to other documents containing the automated test information and the English language readout to the test programs.

c. A backup signal flow schematic diagram of the system or component under test shall be used to support troubleshooting if automatic test equipment is not available or fails. The schematic shall show all test points and input and output signals.

d. Whenever systems have primary diagnostics utilizing semiautomatic or automatic test equipment, manual or alternate troubleshooting/diagnostic procedures shall be developed when specified by the contracting activity.

5.3.9.4.3 Procedures for built-in test (BIT) using preprogrammed electronic imbedded media. Testing procedures shall identify the specific part number (e.g., ICD, UUT, TPS, etc.) required for test performance. The illustrations shall depict how the program works and the primary objectives of the system test. Procedures shall normally include, but are not limited to, the following:

a. Identification of any requirements for controlled environmental conditions during transportation, storage, and use of equipment or material (e.g., controlled temperature, humidity, or clean room facilities for computers and preprogrammed electronic imbedded media).

b. Identification of any pretest requirements.

c. Step-by-step instructions covering the installation of the electronic imbedded media within the system and any precautions or observations to be noted during this process.

d. Test procedures (e.g., system turn on, identification of time required to run and complete the system test, an indication of any possible mid-test interruptions or stoppages and how to respond to them).

e. Testing setup procedure and troubleshooting criteria defined in separate TMs shall not be duplicated.

f. Reference shall be made to other documents containing automated test information and the English language readout of the test program.

g. The same type backup information is required as identified in 5.3.9.4.2.

h. The troubleshooting information shall be accompanied by logic charts, schematics, signal flow diagrams, tables,

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and other illustrations as required for comprehensive understanding of the procedures.

5.4 Quality assurance requirements. Quality assurance and quality control for TMs, revisions, supplements, and changes shall be in accordance with MIL-STD-361-4.

5.5 Packaging and preparation for delivery. TMs, revisions, supplements, and changes shall be packaged in accordance with MIL-STD-361-5.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document prescribes the troubleshooting requirements applicable to various types of technical manuals, revisions, supplements, and changes prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.1.1.)

6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with MIL-M-63008.

6.4 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced. (See 2.1.1.)

c. Environmental control data (4.2.2).

6.5 Supersession data. This document consolidates, standardizes, and supersedes the troubleshooting requirements contained in the following documents:

MIL-M-63036C(TM) - Manuals, Technical: Operator's, Preparation of.

MIL-M-63038C(TM) - Manuals, Technical: Organizational or Aviation Unit, Direct Support or Aviation Intermediate, and General Support Maintenance.

6.6 Subject term (key word) listing. The following terms are to be used to identify this document during retrieval searches:

Built-in test equipment (BITE)
Cable fabrication
Development of text and graphics
Environmental control data
Fault isolation requirements
Logic trees
Major troubleshooting divisions
MAPS list preparation
MAPS list requirements
Measurements and tolerances

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Preparation of illustrations
Quality assurance requirements
Text content
Troubleshooting data
Troubleshooting diagrams
Troubleshooting methods
Troubleshooting procedures

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SYMPTOM INDEX

	Troubleshooting Procedure (Paragraph)
COOLING SYSTEM	
Radiator	
Boils over	3.12
Leaks	3.17
Temperature Gage	
No indication	3.20
Runs cold	2.11
Runs hot	3.13
ENGINE	
Misses	3.34
Overheats	3.72
Won't start	3.27
EXHAUST SYSTEM	
Excessive smoke	3.33
Water vapor	3.33

FIGURE 1. MAPS list sample (for simple- to medium-complex systems) (reference 5.3.6.1, 5.3.6.2).

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CODE	DESCRIPTION	FAULT ISOLATION PROCEDURE
DM-1	080 unit target marker circle range incorrect- radar dominant	3.12
DM-2	080 unit target marker circle elevation incorrect - IR dominant	3.12
DN	□ TSD target bug range and HSI target range incorrect - data link	8.17
DN-1	□ TSD target bug bearing and HSI target bearing incorrect- data link	9.3
DN-2	□ HSI miles indicator incorrect - data link	8.17
DT-5	□ AMI command Mach incorrect - manual	8.10
	DIGITAL COMPUTER	
EA	Computer malfunction - can be cleared	6.3
EA-1	Computer ASIA information incorrect	2.10
EA-2	Computer value of WSEM parameter/CCM signals incorrect	2.4
EA-3	Temperature callup incorrect on AVVI or 980 unit	...
EB	Computer malfunction - cannot be cleared	6.4
EB-1	Computer manual steering erroneous - all modes	3.16
EB-2	Computer manual and AFCS steering erroneous - all modes	8.5
	ARTIFICIAL HORIZON	
FA	Artificial horizon - no erection - roll	8.2
FA-1	Artificial horizon - no erection - pitch	8.2
FC	Artificial horizon - drifts/tumbles - pitch	8.1
FD	Artificial horizon - incorrect erection - pitch	8.2
FJ	Artificial horizon - missing	3.12
FJ-1	Artificial horizon - distorted	3.12
FK	905 unit annunciator incorrect	8.4
	RADAR SEARCH AND DETECTION	
GA	B-sweep missing - on and stby	3.1
GA-1	B-sweep missing - on	3.1
GA-2	B-sweep missing - stby	3.1
GB	B-sweep intensity incorrect	3.2
GC	B-sweep super search - inoperative	3.3
GC-1	B-sweep super search width incorrect	3.3

FIGURE 2. MAPS list sample for medium- to heavy-complex systems) (reference 5.3.6.1, 5.3.6.2).

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ITEM	ACTION	NORMAL INDICATION	CORRECTIVE ACTION
g.	Calculate card extender noise by subtracting DVM (5A2) indication in step 2.f. from DVM (5A2) indication in step 35.d. of figure 7-2.		
h.	On WRA, adjust A15R33 for normal indication.	DVM (5A2) displays -743 +49 mV plus card extender noise.	Replace SRA A15, if indication not normal.
3.	High Threshold Adjustment 137 (A4R34, A4R80)		
a.	Perform item 1 of figure 7-5 to remove top cover from WRA.		
b.	Press and release WRA POWER ON pushbutton.	WRA POWER ON pushbutton lights.	. . .
c.	Press and release FUNCTIONAL TEST/FAULT ISOLATION pushbutton.	FAULT ISOLATION portion of pushbutton lights.	. . .
d.	Press and release DVM pushbutton.	DVM pushbutton lights.	. . .
e.	Select signal number 011.	DVM (5A2) displays 110 +10 mV.	PASS: Step 3.m. FAIL: Step 3.f.
f.	Record DVM (5A2) indication for use in step 3.k.		
g.	Press and release WRA POWER OFF pushbutton.	WRA POWER OFF pushbutton lights.	. . .
h.	On WRA, extend SRA A4 using card extender (3070702-3).	. . .	. . .
i.	Press and release WRA POWER ON pushbutton.	WRA POWER ON pushbutton lights.	. . .
j.	Record DVM (5A2) indication for use in step 3.k.	. . .	. . .

FIGURE 3. Example of troubleshooting integrated into maintenance (alignment) procedure (reference 5.3.7.1).

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ITEM	ACTION	NORMAL INDICATION	TROUBLESHOOTING REF
15.	Display Test 005		
	a. On WRA, perform the following:	. . .	. . .
	(1) Set ASPECT switch (S7) to TAIL.		
	(2) Set MLC switch (S8) to AUTO.		
	(3) Set XMTR CHAN thumb-wheel (S9A) to 2.		
	(4) Set MSL CHAN thumb-wheel (S9B) to 3.		
	b. Select test number 005.	TEST IN PROGRESS readout displays 005.	. . .
	c. On WRA, press and hold SPL (S5A) and ALT DIF (S5B) pushbuttons.	. . .	. . .
	d. Press and release TEST EXECUTE pushbutton twice.	TEST RESULT readout displays NO GO and SSI BIT FAULT 29 and 30 indicators light (-165 and up configuration WRAs) or TEST RESULT readout displays GO and all SSI BIT FAULT indicators are off (-155 thru -161 configuration WRAs).	3
		On WRA, IROT indicator (A1DS4) is lit.	30
		On WRA, SPL (S5A) and RDR (S11A) pushbutton light green.	32
		On WRA, RANGE readout (A1M1) displays 50.	21
		On WRA, WCS MODE readout (M3) displays MRL.	22
		TEST STATUS E indicator is lit.	29
		TEST STATUS FF indicator is lit.	33
		TEST STATUS BB indicator is off.	24

FIGURE 4. Example of user referred to separate troubleshooting data (sample 1) (reference 5.3.7.2).

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d. Sequence 3 (which may reference sequences 5 through 8).

e. Sequence 4 (which may reference sequence 6).

WARNING

On hcu (figure 21, detail B) ensure WCS switch (7) is set to OFF when disconnecting or connecting electrical connectors and when removing or replacing units, or injury or death to personnel may result.

NOTE

Ensure that pretest setup procedures (para 4.2) have been performed before proceeding.

44. Power Turn-on.

45. To perform WCS turn-on, proceed as follows:

a. On hcu (figure 21, detail B) set WCS switch (7) to STBY and check that power indicator comes on in 30+ seconds. _____ TN000

NOTE

As a function of the tactical program, the computer test (sequence 2) is performed automatically during the 30-second displays time-in. At the successful completion of sequence 2 a normal BIT presentation with a pass symbol ( ) is displayed for approximately 10 seconds. At this time the tid switches to a tactical display. If sequence 2 fails, an fi display appears. This display may be removed by pressing PRGM RESTRT pushbutton on CAP. A manual sequence 2 should be run if the automatic sequence 2 fails.

b. Check that power reset indicator remains out. _____ TN0005

c. Press and hold LIGHT TEST switch and check that proper indicators (figure 8) come on.

FIGURE 5. Example of user referred to separate troubleshooting data (sample 2) (reference 5.3.7.2).

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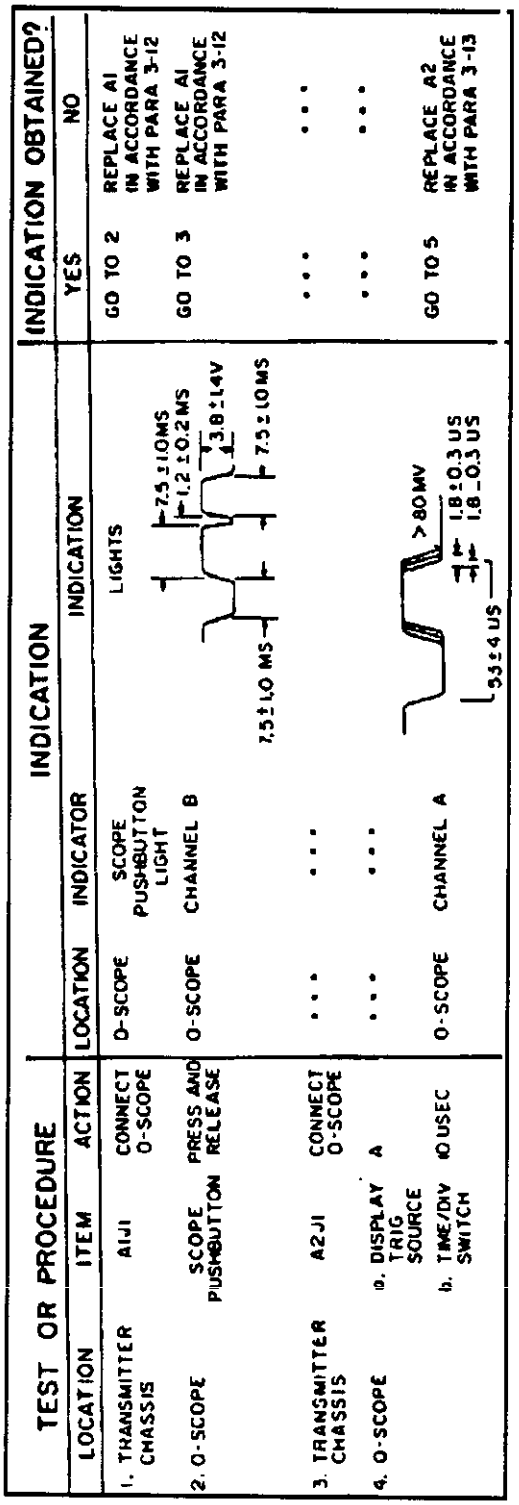


FIGURE 6. Example 1 of tabular logic tree (reference 5.3.8.1.3).

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STEP	INSTRUCTION	INDICATION	YES	NO	REMARKS
	WARNING In the following procedure, operation of the armament bay doors and missile launchers is a potential safety hazard to personnel in the vicinity of the aircraft. Adhere to all safety precautions (TM X-XXXX-XXX-XX).				
1	(TM 27-2) ATS Check: Perform	(Master warning panel) MISFIRE: Goes off when TEST MISFIRE FWD and TEST MISFIRE AFT are set to down (off).	2	(054) (Launcher harness) (596) (Intervalometer) (Misfire relay)	MISFIRE remains on after the fire signal if either a ground path is maintained to the misfire relay hold-in coil or +28 v is maintained to the misfire relay pull.
	NOTE Perform steps 2 and 3 on suspected faulty rail only, or all rails for a complete system check.				
2	(054) Missile gone switch: Actuate to the up position for entire step. (Missile simulator) MISSILE SIMULATE: GONE (27-2) ATS Check: Perform with S96 TEST MISFIRE switches in the down (off) position.	(Master warning panel) MISFIRE: Blinks on momentarily when the fire signal is delivered.	3	(054) Launcher harness - - - - -	Missile gone switch circuit in launcher may be faulty.
3	(054) Missile gone switch: Actuate to the up position for entire step. (Missile simulator) MISSILE SIMULATE: PRESENT (TM 27-2) ATS Check: Perform.	(Master warning panel) MISFIRE: Remains on after the fire signal is delivered. (Armament bay) Doors: Remain open. Launchers: Remain extended.	▲	(054) (Launcher harness) - - - - -	Checks the ability of the system to stop the armament sequence when a misfire condition is simulated.

FIGURE 7. Example 2 of tabular logic tree (reference 5.3.8.1.3).

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8.59 B-SWEEP OR EL MARKER JUMPS SLIGHTLY WHEN ACTION SWITCH IS RELEASED.

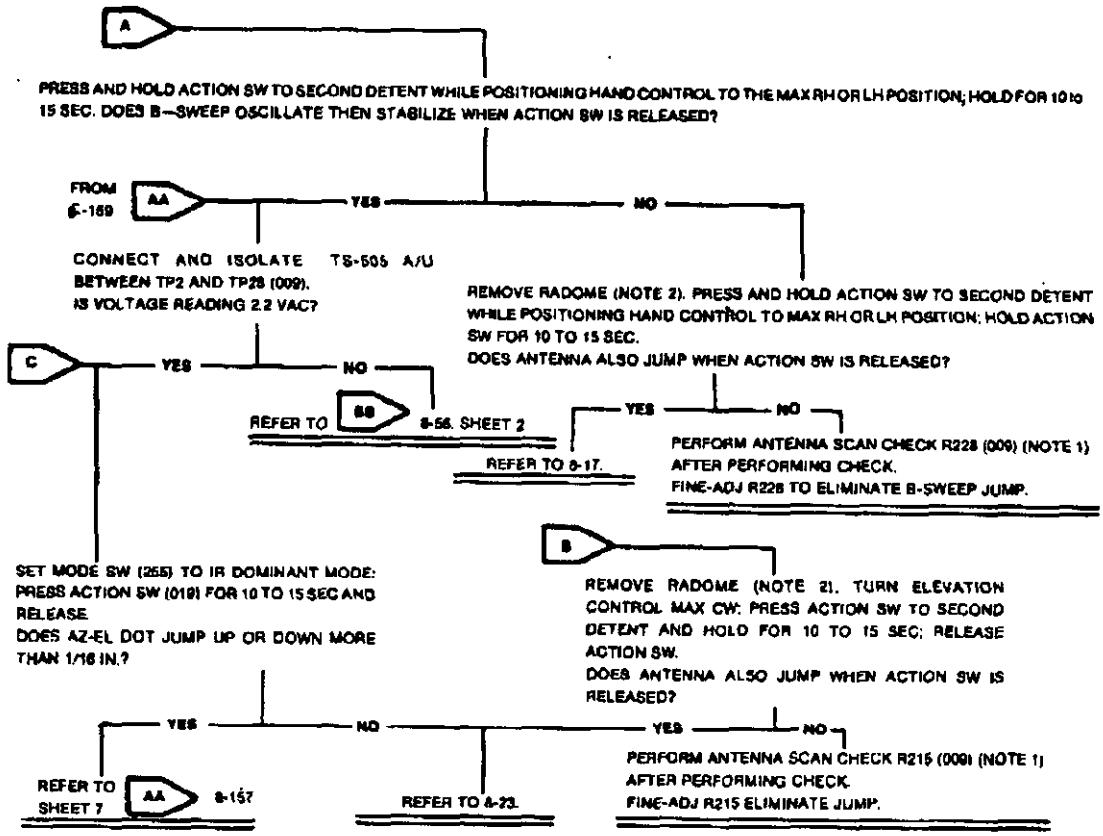


FIGURE 8. Example 1 of branching logic tree (reference 5.3.8.1.3).

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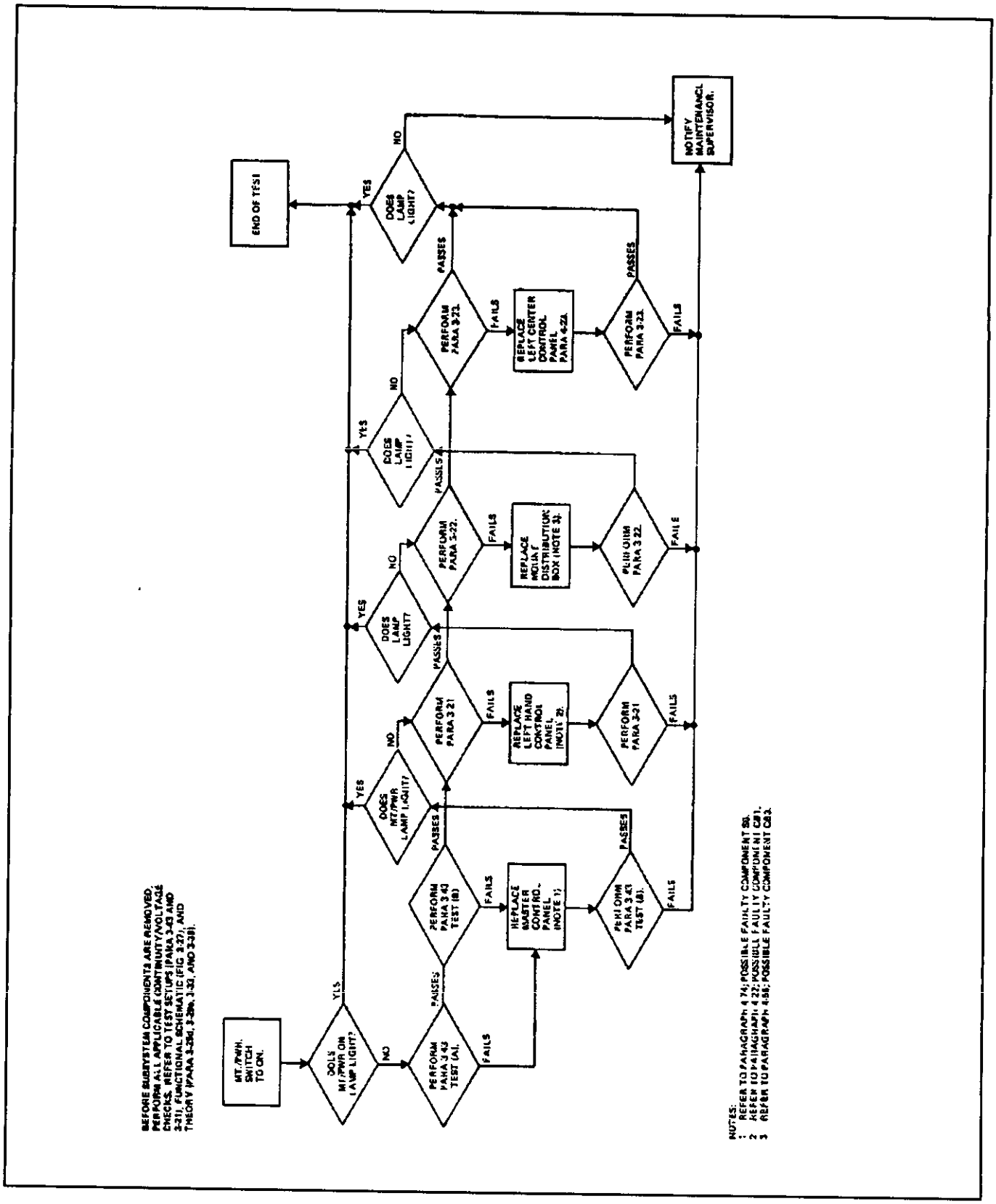


FIGURE 9. Example 2 of branching logic tree (reference 5.3.8.1.3).

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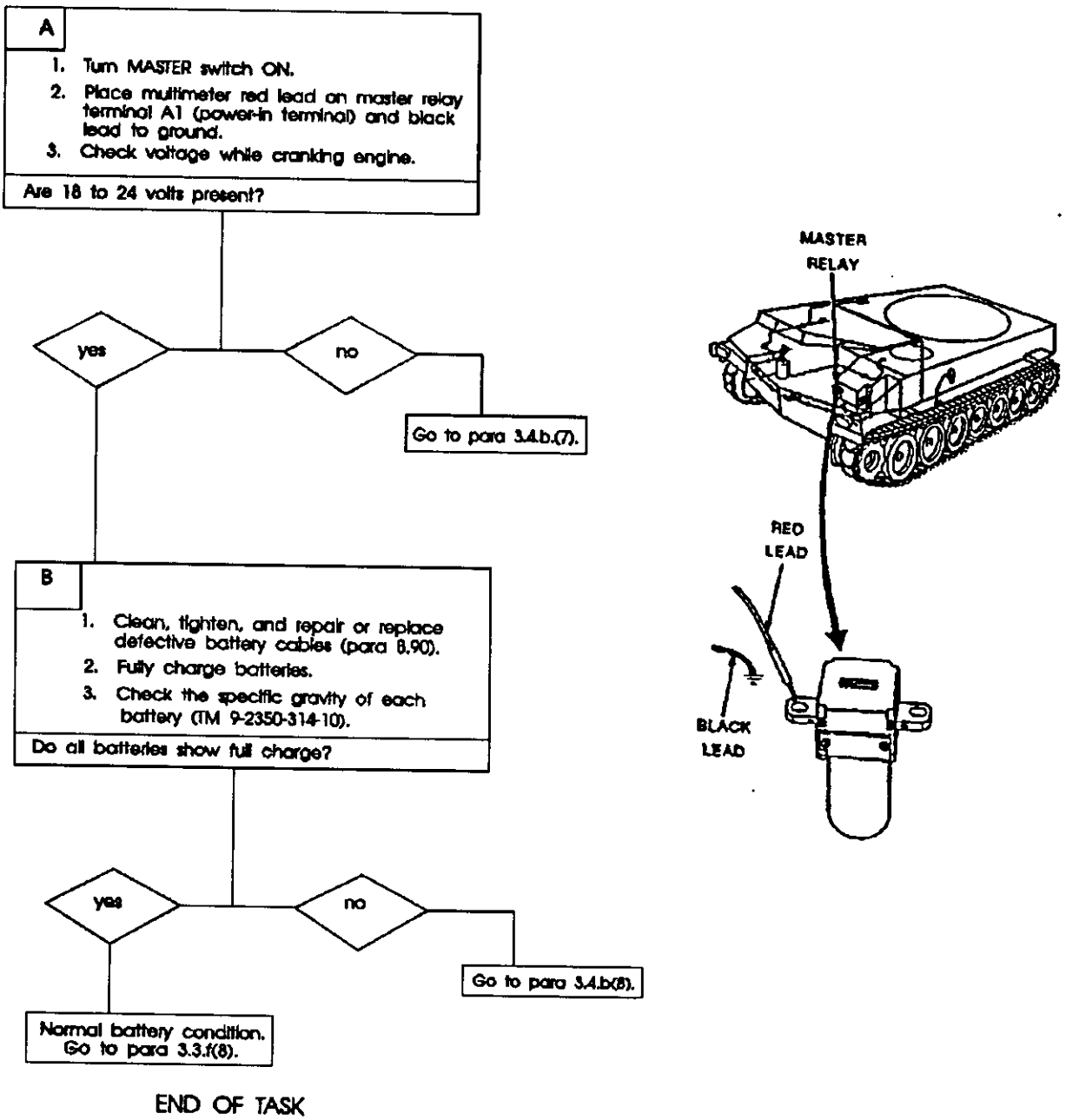


FIGURE 10. Example 3 of branching logic tree (reference 5.3.8.1.3).

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MALFUNCTION
TEST OR INSPECTION
CORRECTIVE ACTION

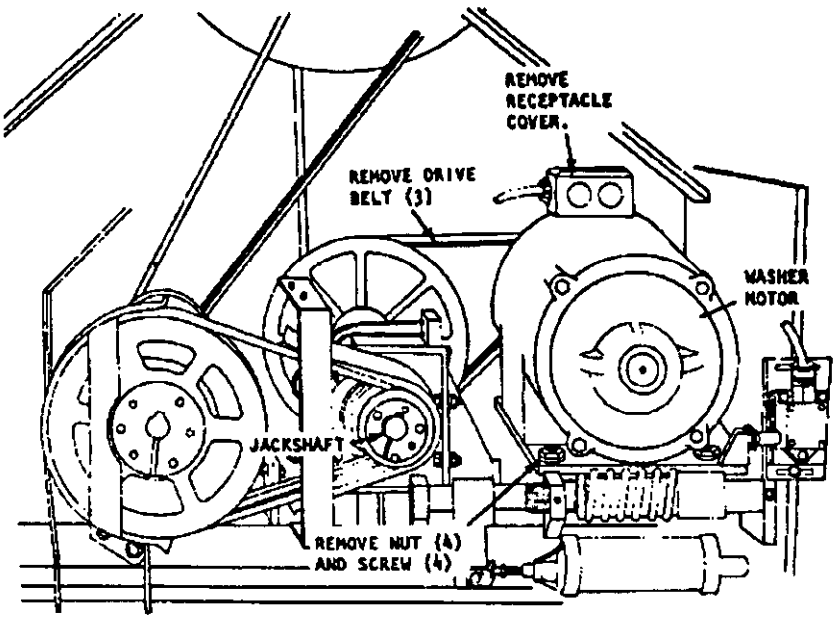
16. WATER PUMP FAILS TO ROTATE.

Step 1. Check for a broken or cracked motor, damaged shaft threads, and bent shaft.
Replace motor, if defective. Remove the motor using illustrations and instructions listed in figures 4-11 and 4-12.
Step 2. Check for a clogged impeller.
Clean the impeller (para 2.10).

17. WASHER-EXTRACTOR BASKET WILL NOT ROTATE.

Step 1. Check for locked brake assembly.
Correct pressure or replace possible defective air cylinder (fig. 4-95).
Step 2. Check for a broken or cracked motor, damaged shaft threads, and bent shaft.
Refer to figure 4-13 (Eidal Model) and figure 4-14 (Edro Model) and replace motors.

NOTE: Tag and disconnect electrical leads as necessary.



NOTE: For removal of drive belts only, loosen nut (4) and slide washer motor toward jackshaft. Remove other drive belts in a similar manner.

FIGURE 11. Example 1 of narrative logic tree (reference 5.3.8.1.3).

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Table 4-19. Receiving System Quarterly Check Procedures - Continued.

STEP	OPERATION └→NORMAL INDICATION └→ADJUSTMENTS/REFERENCES
10.	**CHECK 22 DB STC (R23) - Continued**
f.	(CMST) TARGET CHANNEL ATTENUATION: Increase setting by 1 dB. └→ (Control indicator) Paint level in range gate 2 decreases. └→ Repeat step e. above.
11.	**CHECK 18 DB STC (R22)**
a.	(CMST) TARGET CHANNEL ATTENUATION: Set 18 dB lower than the setting recorded in step 10c. └→ (Control indicator) Paint level in range gate 3 is approximately 80 percent. ▼ A3A2A1A1R22 (see step 9). If paint level is 100 percent, adjust R22 slowly cw until paint level decreases below 80 percent, then slowly ccw until the paint level is approximately 80 percent, or suddenly jumps to 100 percent. If the paint level is much less than 80 percent, adjust R22 slowly ccw until the paint level is approximately 80 percent or suddenly jumps to 100 percent.
b.	(CMST) TARGET CHANNEL ATTENUATION: Increase setting by 1 dB. └→ (Control indicator) Paint level in range gate 3 decreases. └→Repeat step a. above.

FIGURE 12. Example 2 of narrative logic tree (reference 5.3.8.1.3).

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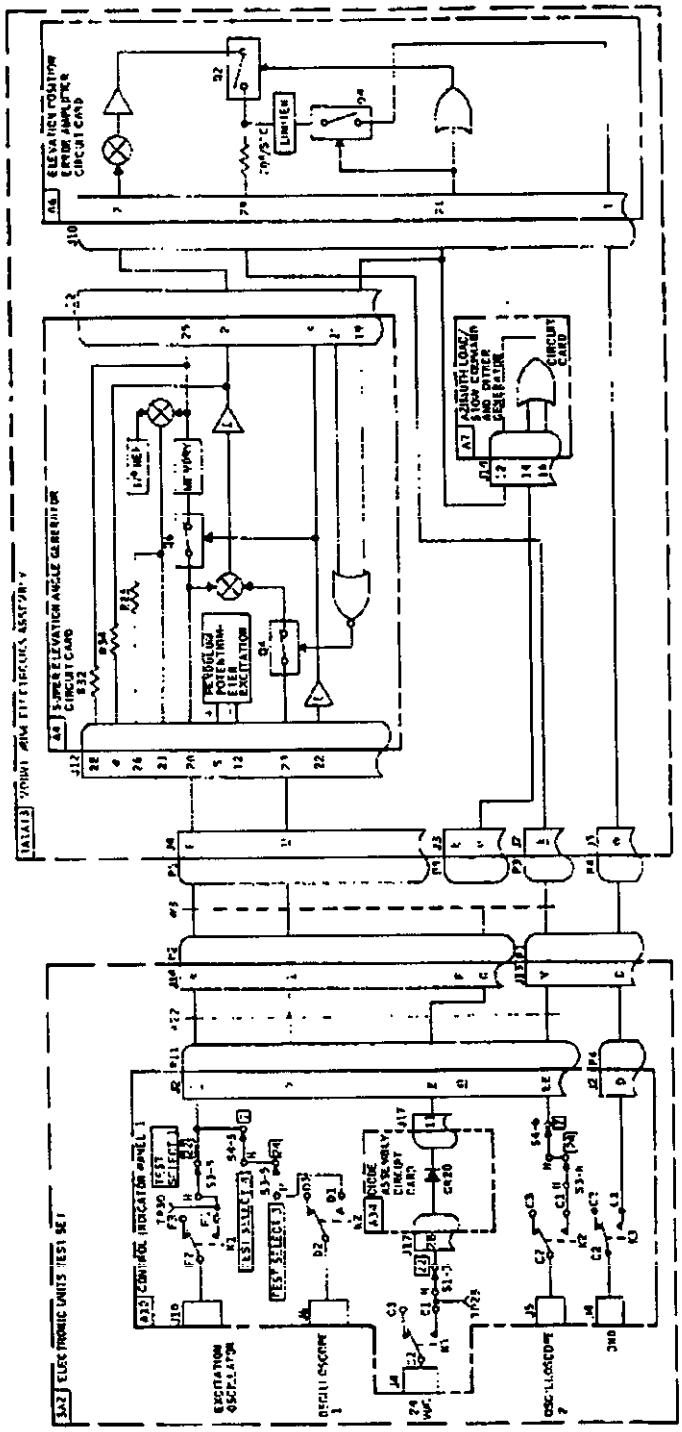


FIGURE 13. Example of test loop functional diagram (reference 5.3.8.2.1b).

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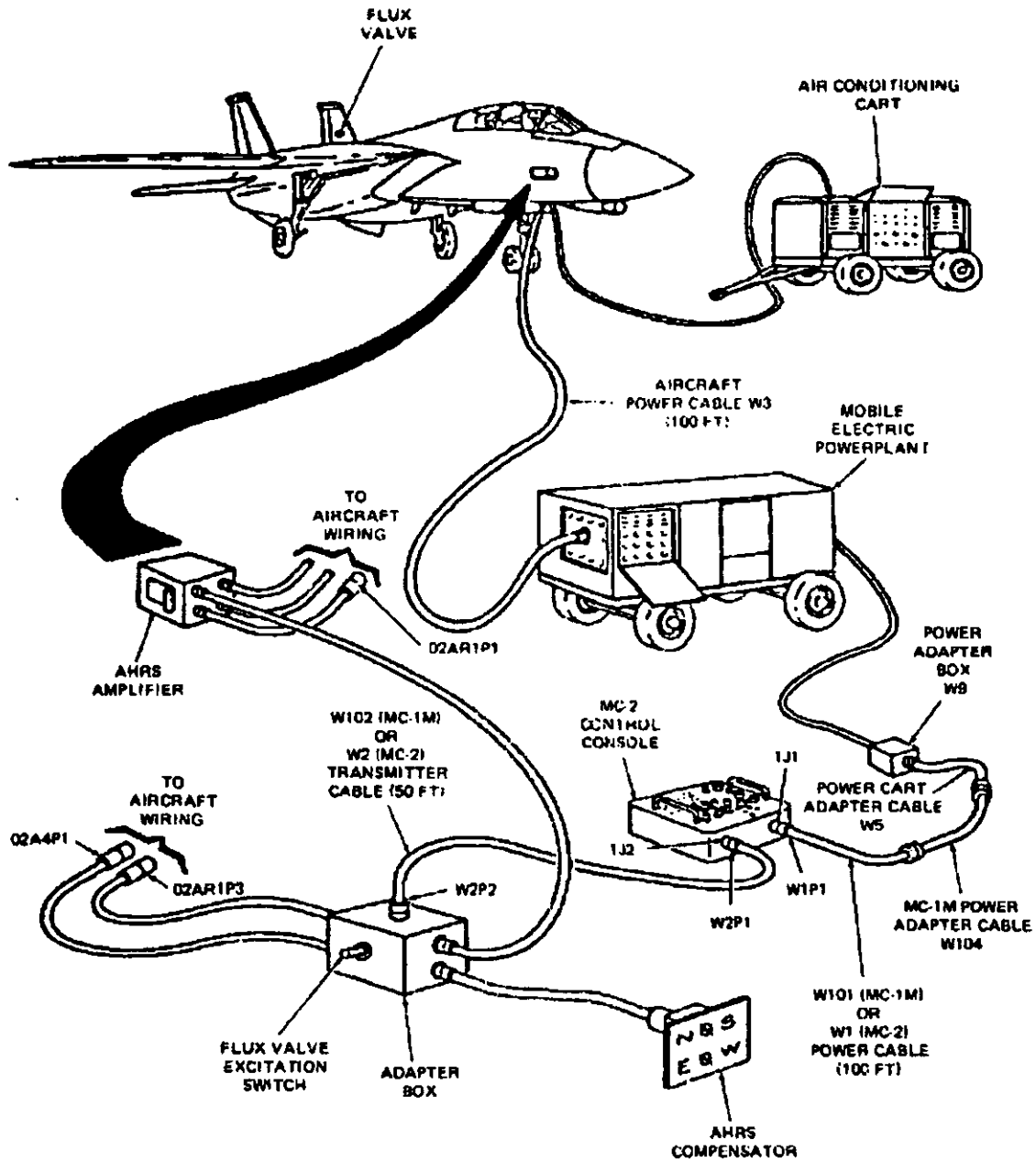


Figure 6-2. Compass System Test Setup.

FIGURE 14. Example of testing setup illustration for testing and troubleshooting procedures (reference 5.3.9.1a, 5.3.9.4.1g).

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TM X-XXXX-XXX-XX

5.6 TESTING/TROUBLESHOOTING.

Support Equipment Required:

(Alternate Equipment Part Numbers for Equipment Designation are shown in parenthesis.)

NC8A(NC10)	Mobile Electric Power Plant
1U22883	Adapter Cable W5
NR5C(NR5B)	Air-Conditioning Cart Electric-Driven
MC-2(MC1-M)	Magnetic Compass Calibrator Set
127AH10	Tow Bar
AN/PSM-4D	Multimeter
H-173A/AIC	Headset (3)
M-96A/A	Dynamic Microphone (3)
CX-4620/AR	70-Foot Cordset
ACEL33C	Oxygen Hose Assembly (2)
AM-3697B/A	In-Line Amplifier (2)
127-C-618-1	Cordset (2)
U22708	Adapter Plate 2-inch Nonmagnetic Screwdriver

5.7 PRETEST SETUP.

a. On compass nose, align aircraft so that nose points north.

b. Connect cordset with in-line amplifier, dynamic microphone, headset, and oxygen hose assembly to 08HTA1P1 on left of pilot seat.

c. Connect cordset with in-line amplifier, dynamic microphone, headset, and oxygen hose assembly to 08HTA2P1 on left side of NFO seat.

5-3

FIGURE 15. Example of pretest setup for testing and troubleshooting procedures (reference 5.3.9.1a, 5.3.9.4.1g).

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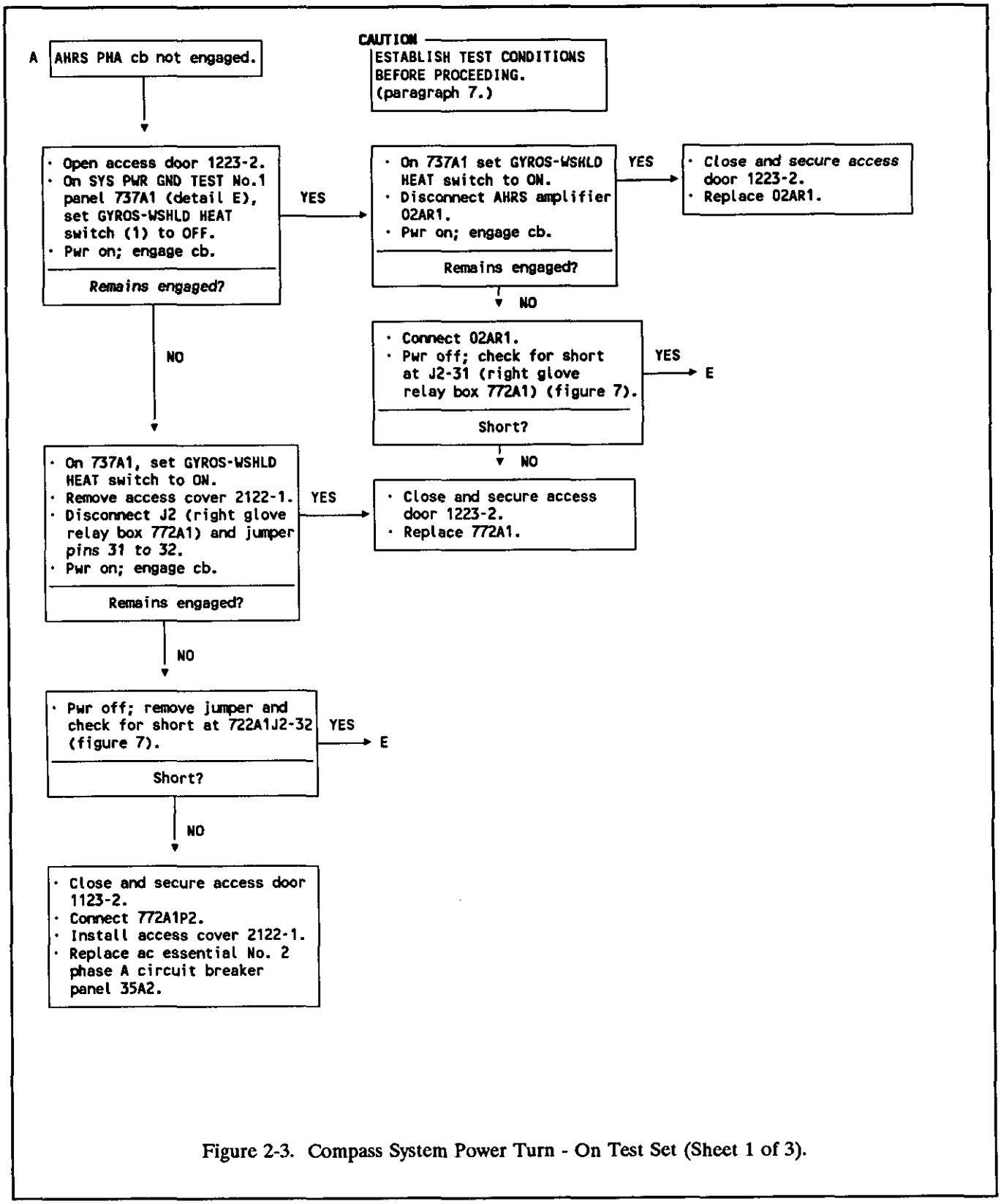


Figure 2-3. Compass System Power Turn - On Test Set (Sheet 1 of 3).

FIGURE 16. Example of functional block diagram for testing and troubleshooting procedures (reference 5.3.9.1d, 5.3.9.4.1g).

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Table 3-1. Recorder/Reproducer Does not Run When Speaking Into Microphone. See figure 3-2.

	NO	YES
a. Apply external electrical power (TM X-XXXX-XXX-XX). Set station box I/C switch to I/C, I/C volume control full clockwise and speak into microphone. Was sidetone heard?	b	c
b. Replace station box (TM X-XXXX-XXX-XX)	-	-
c. Remove electrical power (TM X-XXXX-XXX-XX). Disconnect plug F7H from station box and F7V from recorder. Using AN/USM-311 multimeter, make continuity check between pin 11 of F7H and pin H of F7V. Does continuity exist?	d	e
d. Isolate and repair wiring	-	-
e. Using multimeter, make continuity check between pins E and G of F7V and aircraft ground. Does continuity exist?	d	f
WARNING To prevent injury, air brake safety devices must be properly installed prior to entry into electrical equipment bay.		
f. Disconnect plug C10G from distribution panel. Using multimeter, make continuity check between pin C of C10G and pin A of F7V. Does continuity exist?	d	g
g. Apply battery power on aircraft to no. 2 DC busbar. Using multimeter, check for presence of battery voltage at pin C of receptacle for plug C10G. Is voltage present?	h	i
h. Refer to DC Power Distribution (TM X-XXXX-XXX-XX)	-	-
i. Remove battery power. Connect all connectors. Apply external electrical power (TM X-XXXX-XXX-XX). Perform auto-start adjustment per TM X-XXXX-XXX-XX. Does recorder now run with microphone input?	k	j
j. Perform auto stop delay adjustment (TM X-XXXX-XXX-XX)	-	-
k. Replace recorder	-	-

FIGURE 17. Example of fault logic text for testing and troubleshooting procedures (reference 5.3.9.1e).

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Table 4-8. Troubleshooting - Continued.

FAULT	PROBABLE CAUSE	REMEDY
Excessive Fuel Pressure	a. Restriction in pressure line.	a. Remove restriction or replace pressure line.
	b. Filter elements dirty.	b. Replace.
Pump Overheating	a. Fuel supply low or exhausted.	a. Replenish fuel supply.
	b. Insufficient pumping clearance at impeller.	b. Check for correct impeller installation, or bent shaft; repair or replace.
	c. Extended operation with discharge valve closed.	c. Open discharge valve and allow fuel to circulate until pump cools.
Does Not Pump	Pump casing not full of fluid (not primed).	Be sure pump casing is full of fluid before starting.
Pumps at Reduced Flow	a. Rotation backward.	a. Exchange any 2 of 3 wires connecting motor to power; refer to wiring diagram.
	b. Filter elements dirty.	b. Replace.
	c. Air entrained in pump casing (not completely primed).	c. Bleed out air at bleed valve at top of casing while running.
	d. Mechanical seal leaking air into pump. (Likely to happen at higher vacuum conditions at inlet.)	d. Replace seal.
	e. Low line voltage or motor connected to wrong voltage source. (Motor not maintaining speed.)	e. Correct voltage condition or connections.
	f. Excess clearance between wear ring and impeller.	f. Replace wear ring and/or impeller as required.

FIGURE 18. Example of a table for testing and troubleshooting procedures (reference 5.3.9.1e, 5.3.9.4.1g).

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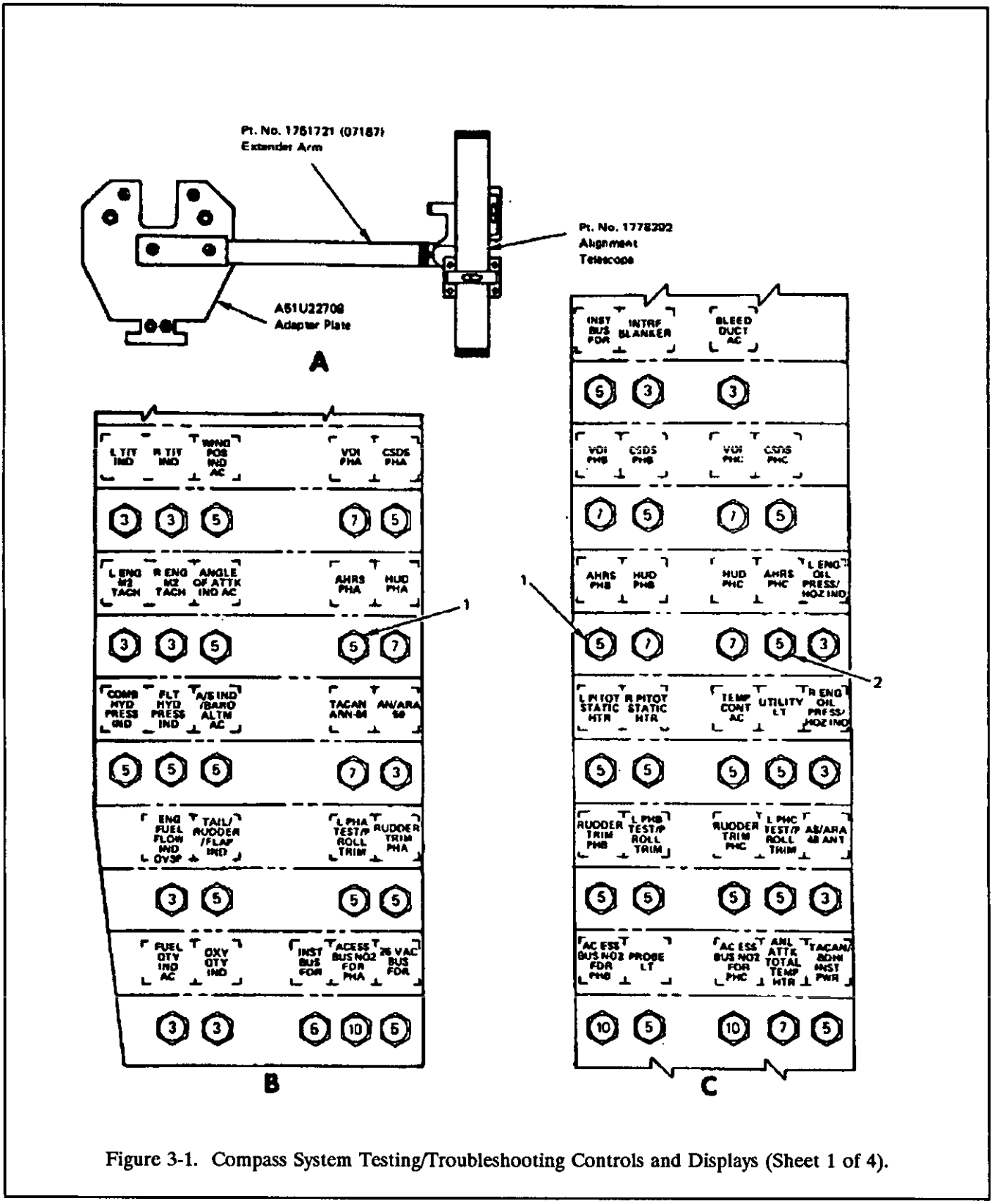


Figure 3-1. Compass System Testing/Troubleshooting Controls and Displays (Sheet 1 of 4).

FIGURE 19. Example of controls and displays illustration for testing and troubleshooting procedures (reference 5.3.9.4.1f).

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3. DOCUMENT TITLE
Troubleshooting Procedures

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
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